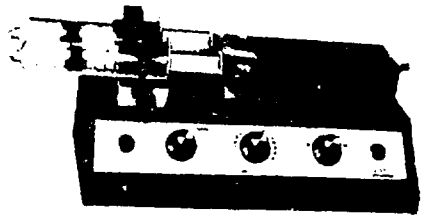
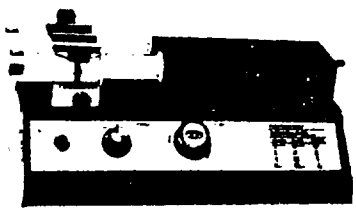


MS36590-1



MS36590-2



MS36590-3

PART NUMBER	SYRINGE CAPACITY	FLOW RATE	NUMBER OF SYRINGES
MS36590-1	10 µL to 50 mL	0.01 µL/hr to 13 mL/min	1
MS36590-2	5 mL to 100 mL	0.15 mL/min to 60 mL/min	3 up to 30 cc 2 up to 50 cc 1 - 100 cc
MS36590-3	0.5 µL to 100 mL	0.07 µL/day to 140 mL/min	3 up to 30 cc 2 up to 50 cc 1 - 100 cc

MATERIAL: Housing shall be made of vinyl coated aluminum. All other materials shall be as required for satisfactory functioning.

DESIGN: Pumps shall meet the following design requirements:

MS36590-1 shall hold a single pump for continuous infusion and shall hold all types of syringes in any size up to 50 mL. Pumps shall be precalibrated and highly reproducible with eleven rates in each range and a total of 22 for each syringe size. Pulser motor shall provide power through set of gears moving syringe piston smoothly down its barrel. Motor shall stall at end of run to prevent syringe damage.

MS36590-2 pump shall have direct dialing of 19 most frequently used flow rates and is precalibrated to infuse or withdraw fluids, only one syringe can be held in holder for withdrawal mode. Gear drive shall provide a continuous and uniform flow at all except slowest rates with largest syringes. Slip clutch shall protect driver motor and syringe at end of travel.

USER ACTIVITIES: ARMY - MD, DM
NAVY - MS

REVIEWER ACTIVITIES: ARMY - ME
NAVY - OS
AIR FORCE - 82

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

APPROVED 20 JUL 1983 REVISED

P.A. EA	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
Other Cust MS 99		PUMP, SYRINGE	MS36590
Procurement Specifications NONE	SUPERSEDES:		SHEET 1 OF 2

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MS36590-3 pump shall be continuously variable with complete range covered by direct dialing and no gear changes. Syringe holder shall meet all pumping needs, accepting syringes of any type including microliter. Pump drive shall be geared to maintain uniform flow at all speeds, with built-in slip clutch protection. Electronic feedback circuit controls reproducibility regardless of back pressure variations. Stabilized DC motor shall drive syringe piston for uniform fluid flow.

Syringe pumps shall be enclosed to prevent damage from solutions and mechanism shall allow for quick set-up for filling. Units shall operate on 115/120V AC, 50/60Hz.

All electrical components shall conform to the standards of Underwriters Laboratories, Incorporated.

NOTE: Items differing from this Military Standard by minor variations in design or dimensions are considered standard under this document, provided such variations do not affect performance and are not in conflict with the material requirements specified herein.

FSCM: 13134 and 59728.

USER ACTIVITIES:

REVIEWER ACTIVITIES:

APPROVED REVISED

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. Other Cust	INTERNATIONAL INTEREST	TITLE PUMP, SYRINGE	MILITARY STANDARD MS36590
Procurement Specifications	SUPERSEDES:		SHEET 2 OF 2

DD FORM 1 MAY 73 672 (COORDINATED) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

PROJECT NUMBER 6640-1189

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